

Jason L. Fu

San Jose, CA

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EDUCATION

Cornell University, College of Engineering

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

GPA: 3.92/4.00

Ithaca, NY

Aug 2023 - May 2027

EXPERIENCE

Software Engineering Internship

Fremont, CA

HYC USA

Jun 2025 - Aug 2025

- Engineered the backend architecture for a warehouse management system using FastAPI and SQLite, implementing concurrent programming paradigms to prevent race conditions and ensure highly available, real-time data synchronization for 50+ enterprise users.
- Designed secure, high-throughput RESTful APIs with JWT authentication and RBAC, and managed zero-downtime schema evolution via Alembic database migrations to ensure strict system reliability across automated CI/CD deployments.
- Architected optimized backend search algorithms and streamlined CI/CD pipelines, reducing database retrieval latency by 30% to support seamless, high-volume operations for global teams.

Software Engineering Internship

Fremont, CA

HYC USA

Jun 2024 - Aug 2024

- Integrated Meta Llama 3.1 language models (LLMs) in in-circuit testing automation using Python GUI, PyTorch, and Transformers with speech recognition and text-to-speech capabilities for streamlined workflows.
- Partnered with hardware engineers to design a PyQt5-based testing GUI for Apple, Tesla, and Meta circuits, reducing test cycle time by 15% through optimizing data handling and automation.

PROJECTS

Structured Textual Data Ingestion Pipeline

Ithaca, NY

Cornell AguaClara

Feb 2025-Present

- Designed and implemented an AWS-powered data pipeline (Bedrock, S3, Lambda, API Gateway, OpenSearch, Cognito) enabling 40GB+ of research data to be searchable, improving query efficiency for 50+ researchers.
- Architected a fault-tolerant, decoupled distributed system using AWS SQS and Step Functions for automated infrastructure provisioning, managing high-throughput data pipelines and reducing deployment time by 40%.
- Implemented RESTful APIs with AWS Lambda to support retrieval-augmented generation (RAG), enhancing accuracy and context-awareness of LLM-driven research queries.

Plant Operations Smartphone Tracker | <https://github.com/AguaClara/aguadatos-amplify-android>

Ithaca, NY

Cornell AguaClara

Jan 2024-May 2025

- Designed and launched POST, an Android app (Flutter, Kotlin, Jetpack) enabling operators in Central America and India to monitor water filtration data, improving operational efficiency for plants serving 100K+ people.
- Engineered an offline-first distributed backend using AWS AppSync (GraphQL) and DynamoDB, implementing state management and conflict resolution protocols to guarantee data consistency upon network reconnection.
- Contributed to the open-source Cornell AguaClara repository (50+ active contributors) by developing a responsive frontend with Kotlin, XML, and Jetpack architecture components to streamline operator workflows.

EduConnect.AI | <https://github.com/jlfud/ai-hackathon-educonnect>

New York, NY

Hackathons at Cornell University

Feb 2025-May 2025

- Collaborated on a full-stack web app using React and Streamlit (Python) to connect students with nonprofits, integrating AI-driven interfaces that streamlined matching and improved user engagement.
- Prototyped an AI-powered matching algorithm using OpenAI API and SerpAPI (RAG), improving recommendation relevance and discovery accuracy for nonprofit partnerships.

TECHNICAL SKILLS

- Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, Kotlin, OCaml, HTML/CSS
- Frameworks/Libraries:** React, Node.js, Express, FastAPI, Flask, MongoDB, GraphQL, Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers
- Cloud & Architecture:** AWS (Lambda, S3, DynamoDB, Amplify, CloudFormation/CDK, Cognito, Step Functions, SQS), Docker, Microservices, RESTful APIs, CI/CD, Distributed Systems, JWT, RBAC
- Developer Tools & Testing:** Git, Postman, pytest, JUnit, Alembic, SQLite, OpenAI API, SerpAPI